Question:

Let \(O=(0,0)\), \(A=\left(\tfrac{1}{2},0\right)\), and \(B=\left(0,\tfrac{\sqrt{3}}{2}\right)\) be points in the coordinate plane. Let \(\mathcal{F}\) be the family of segments \(\overline{PQ}\) of unit length lying in the first quadrant with \(P\) on the \(x\)-axis and \(Q\) on the \(y\)-axis. There is a unique point \(C\) on \(\overline{AB}\), distinct from \(A\) and \(B\), that does not belong to any segment from \(\mathcal{F}\) other than \(\overline{AB}\). Then \(OC^2=\tfrac{p}{q}\), where \(p\) and \(q\) are relatively prime positive integers. Find \(p+q\).

Reasoning + Answer:

<think> To find the point \(C\) on \(\overline{AB}\) that does not belong to any segment from \(\mathcal{F}\), we first need to understand the geometry of the problem. The line segment \(\overline{AB}\) passes through the points \(A = \left(\frac{1}{2}, 0\right)\) and \(B = \left(0, \frac{\sqrt{3}}{2}\right)\). The equation of the line \(\overline{AB}\) can be derived as follows:

The slope of \(\overline{AB}\) is:
\[
\text{slope} = \frac{\frac{\sqrt{3}}{2} - 0}{0 - \frac{1}{2}} = -\sqrt{3}
\]
Using the point-slope form of the line equation \(y - y_1 = m(x - x_1)\) with point \(A = \left(\frac{1}{2}, 0\right)\), we get:
\[
y = -\sqrt{3}\left(x - \frac{1}{2}\right)
\]
Simplifying, the equation of the line \(\overline{AB}\) is:
\[
y = -\sqrt{3}x + \frac{\sqrt{3}}{2}
\]
To find the point \(C\) on \(\overline{AB}\) that is distinct from \(A\) and \(B\), we need to consider the conditions under which a segment \(\overline{PQ}\) with \(P\) on the \(x\)-axis and \(Q\) on the \(y\)-axis can intersect \(\overline{AB}\). The length of \(\overline{PQ}\) is 1, and we need to find the point \(C\) such that no segment \(\overline{PQ}\) with unit length can intersect \(\overline{AB}\) except at \(C\).

The point \(C\) can be found by ensuring that the segment \(\overline{AB}\) has a unique intersection with the segments of \(\mathcal{F}\). This unique intersection occurs at the point where the line through \(A\) and \(B\) intersects the circle with radius 1 centered at the origin. The circle's equation is \(x^2 + y^2 = 1\). The line \(\overline{AB}\) has the equation \(y = -\sqrt{3}x + \frac{\sqrt{3}}{2}\). We solve the system of equations:
\[
x^2 + y^2 = 1 \quad \text{and} \quad y = -\sqrt{3}x + \frac{\sqrt{3}}{2}
\]
Substituting \(y = -\sqrt{3}x + \frac{\sqrt{3}}{2}\) into \(x^2 + y^2 = 1\), we get:
\[
x^2 + \left(-\sqrt{3}x + \frac{\sqrt{3}}{2}\right)^2 = 1
\]
\[
x^2 + 3x^2 - 3x\sqrt{3} + \frac{3}{4} = 1
\]
\[
4x^2 - 3x\sqrt{3} - \frac{1}{4} = 0
\]
Using the quadratic formula \(x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\) with \(a = 4\), \(b = -3\sqrt{3}\), and \(c = -\frac{1}{4}\):
\[
x = \frac{3\sqrt{3} \pm \sqrt{(3\sqrt{3})^2